

Lecture 1: Introduction and Dynamic Modelling

Computational Methods for Heterogeneous-Agent Macro

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Model Structure

These slides: solving heterogeneous-agent (HA) GE models with and without aggregate shocks in Julia.

We cover discrete-time GE models with the following form:

1. A forward-looking household block made up of several **Stages**
2. A set of equations describing the rest of the economy, such as firms, government, etc., typically represented as a **directed acyclic graph** (DAG) of functions.
3. A set of **outer loops** solving, e.g., equilibrium, steady state, calibration, transition, or global solution

Course Structure

- Lectures 1-4: Stages.
- Lectures 5-8: Outer loops.
 - Leading up to the “neural Value Function Iteration” (nVFI) global solution algorithm

Motivation

Heterogeneous Agents: Motivation 1

- Empirical average marginal propensity to consume (MPC) high. Average savings **also** high
- Typical representative-agent model cannot easily explain this.
- However, if poor households have high MPC and low saving; and rich households have low MPC/high saving; then a **heterogeneous-agent** model can explain.

Heterogeneous Agents: Motivation 2

- Change in (e.g. interest rate) policy will affect different people **differently**
- A policy will have different **aggregate** effects depending on who it targets.
- We care about the distributional and aggregate consequences of policy.

Household Problem: Two Period

Microfoundation: One Household

- Macroeconomics is the study of how individual behaviour shapes aggregate outcomes.
- To understand the origins of macro outcomes, need to study the sources of individual behaviour.

Two Period Model

Household starts with **wealth** b_0 in Period 0. In Period 1, they receive income z_1 .

$$\begin{aligned} \max_{c_0, c_1 \geq 0} \quad & u(c_0) + \beta u(c_1) \\ \text{s.t.} \quad & b_0 + \frac{z_1}{R} \geq c_0 + \frac{c_1}{R}. \end{aligned}$$

- If you save \$1 today, you can spend $\$R$ tomorrow, but utility discounted by β .
- Lagrangian: $\mathcal{L} = u(c_0) + \beta u(c_1) - \lambda \left(c_0 + \frac{c_1}{R} - b_0 - \frac{z_1}{R} \right)$.

Euler Equation

Lagrangian FOCs:

$$\begin{aligned}u'(c_0) &= \lambda, \\ \beta u'(c_1) &= \frac{\lambda}{R}.\end{aligned}$$

Rearranging,

$$\boxed{\frac{u'(c_0)}{u'(c_1)} = \beta R.}$$

At optimum, **indifferent** between consuming or saving additional marginal unit today for R units tomorrow.

Example: Analytical Solution with Log Utility

Let $u(c) = \log c$. Then, $c_1 = \beta R c_0$, and

$$c_0^* = \frac{b_0 + \frac{z_1}{R}}{1 + \beta}, \quad c_1^* = \frac{\beta R b_0 + \frac{z_1}{R}}{1 + \beta}.$$

With $\beta = 0.96$, $R = 1.04$, $b_0 = 10$, $z_1 = 2$:

$$c_0^* \approx 6.08, \quad c_1^* \approx 6.07.$$

There is **consumption smoothing**. Income is bumpy at $(10, 2)$. Consumption is smoother at $(6.08, 6.07)$.

Policy Function

The **Policy Function** represents optimal period-0 consumption in terms of **initial wealth** b_0 ,

$$c_0^*(b_0) = \frac{b_0 + z_1/R}{1 + \beta}.$$

This way, we have a model of household behaviour, and how **household behaviour** arises.

CRRA Preferences

A standard assumption is constant relative risk aversion (CRRA) utility,

$$u(c) = \begin{cases} \frac{c^{1-\gamma}-1}{1-\gamma}, & \gamma > 0, \gamma \neq 1, \\ \log c, & \gamma = 1. \end{cases}$$

Utility **increasing** in c ,

$$\frac{\partial u(c)}{\partial c} = c^{-\gamma} > 0.$$

Utility is **concave**: Marginal utility **decreasing** in c ,

$$\frac{\partial^2 u(c)}{\partial c^2} = -\gamma c^{-\gamma-1} < 0.$$

Household Problem: T Periods

T Period Model

Lifetime utility is given by,

$$\max_{c_0, \dots, c_{T-1}} \sum_{t=0}^{T-1} \beta^t u(c_t), \quad b_{t+1} = R(b_t - c_t) + z_{t+1}.$$

Solving Lagrangian gives us $T - 1$ Euler equations,

$$u'(c_t) = \beta R u'(c_{t+1}).$$

If future income is perfectly foreseen, can still solve pretty easily.

But what if income is stochastic?

Stochastic Income

- Suppose $\{z_t\}$ follows a Markov chain with transition matrix Π
- Given today's z_t , we know the *conditional distribution* of z_{t+1} , but not the realized value
- In each period t , household chooses c_t without knowing z_{t+1}

Stochastic Euler Equation

With $T = 2$, we can substitute,

$$c_1 = R(b_0 - c_0) + z_1.$$

The problem is just,

$$\max_{c_0} u(c_0) + \beta \mathbb{E} [u(R(b_0 - c_0) + z_1)],$$

with FOC with respect to c_0 ,

$$u'(c_0) = \beta R \mathbb{E} [u'(c_1)].$$

The Euler equation now contains an expectation over *uncertain* c_1 .

Infinite-Horizon Problem ($T = \infty$)

Believe it or not, the problem is actually a little easier if we take $T \rightarrow \infty$:

$$\begin{aligned} \max \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t) \\ \text{s.t. } b_{t+1} = R(b_t - c_t) + z_{t+1}. \end{aligned}$$

But what are we taking $\max$ over? Formally, we need to find a whole sequence of **policy functions**,

$$c_t = c_t(z_0, \dots, z_t) \quad \forall t,$$

but that seems almost impossible. Luckily, we can make the problem easier by reformulating it.
(Note: I'm skipping the issue of the transversality condition.)

Recursive Form

Richard Bellman's First Observation

At each time t , household only makes one choice, then waits,

$$\begin{aligned} \max_{c_t} &= u(c_t) + \beta \mathbb{E}_t \left[\max \mathbb{E}_{t+1} \sum_{s=t+1}^{\infty} \beta^s u(c_s) \right] \\ \text{s.t. } & b_{s+1} = R(b_s - c_s) + z_{s+1} \quad \forall s \geq t. \end{aligned}$$

Importantly, the problem at time t looks **almost** identical to the problem at time $t+1$.

The only possible differences are b_t and z_t . These are the **state variables**.

State Variables

- Informally, **state variables** are those variables which make time periods (and decisions) meaningfully different from each other.
- Here, the state variables are wealth b_t and income z_t . Given these, a household's future conditional expected future utilities and optimal decisions are uniquely determined.

Bellman's Second and Third Observations

Bellman also realized two more things:

2. You can express the first insight as an equation about a **value function** V .
3. If you know this **value function**, you can compute the **policy function** and (numerically) solve the entire model.

In computer science terms, this is an example of **caching**.

Value Function and Bellman Equation

Define the *value function* $V(b, z)$ as the **net present value** of future utility given state (b, z) .

$$V(b, z) = \max_{c \in [0, b]} \mathbb{E}_0 \sum_{s=0}^{\infty} \beta^s u(c_s)$$
$$\text{s.t. } b_{s+1} = R(b_s - c_s) + z_{s+1} \quad \forall s.$$

(Given that all future choices will also be made optimally.)

Then V satisfies the **Bellman equation**

$$V(b, z) = \max_{c \in [0, b]} u(c) + \beta \mathbb{E}[V(b', z') \mid z]$$

$$\text{s.t. } b' = R(b - c) + z'$$

The Value Function Is All You Need

If we know V , then we can compute the **policy function** and solve the whole (household) model,

$$c^*(b, z) = \arg \max_{c \in [0, b]} u(c) + \beta \mathbb{E}[V(R(b - c) + z, z') \mid z].$$

How do we find V ?

- We still have a problem, though. We don't just need to find a number. We need to find an entire **function** V .
- We can do this via “Value Function Iteration” (VFI).
- To understand this: First imagine that we guess a value function, and we get it **wrong**.

The Bellman operator on arbitrary v

Suppose that we guess a **function** v . Even though v might be **wrong**, it would still predict some policy,

$$c^*(b, z; v) = \arg \max_{c \in [0, b]} u(c) + \beta \mathbb{E}[v(R(b - c) + z', z') \mid z].$$

If we use v to **look ahead**, we can get a **different function** $\mathcal{T}v$,

$$(\mathcal{T}v)(b, z) := \max_{c \in [0, b]} u(c) + \beta \mathbb{E}[v(R(b - c) + z', z') \mid z].$$

v kind of represents **beliefs** about the value of being in various states (b, z) .

$\mathcal{T}v$ represents the **updated beliefs** if you reason by looking ahead one period.

The Bellman Operator

- Think of $\mathcal{T}$ as an **operator** that maps a **function** to a **function**: $v \mapsto \mathcal{T}v$.
- This is the **Bellman Operator**.
- The Bellman equation can be rewritten as,

$$V = \mathcal{T}V.$$

- We can solve the household's problem by finding a **fixed point** of $\mathcal{T}$.

Existence and Uniqueness

We will not prove this, but it is useful to note that Banach's fixed point theorem implies that, under mild conditions, there is a **unique** fixed point V of $\mathcal{T}$.

The idea is that $\mathcal{T}$ is a contraction:

$$\|\mathcal{T}v_1 - \mathcal{T}v_2\|_\infty \leq \beta \|v_1 - v_2\|_\infty.$$

Banach implies:

1. A unique fixed point V exists.
2. Iterating $v_{k+1} = \mathcal{T}v_k$ from any starting v_0 converges to V at rate β .

Once More: The Bellman Equation

$$V(\text{state}) = \max_{\text{action}} \text{payoff}(\text{state}, \text{action}) + \beta \mathbb{E}[V(\text{next state}) \mid \text{state}, \text{action}]$$

The rest of this course will basically be variations on this.

Remaining Lectures

Remaining Lectures



Notebooks

Each lecture has a paired Jupyter notebook.

1. Julia 1.10+
<https://julialang.org/downloads/>
2. VS Code + Julia extension.
3. `using Pkg; Pkg.add("IJulia")`.
4. Jupyter (Anaconda or `pip install jupyter`).